

Shadow Tomography Integration with Q-SENSE (PT Method)

Overview

This document describes the integration of **classical shadow tomography** into the Q-SENSE Perturbation Theory (PT) pipeline for estimating Hamiltonian matrix elements. The implementation replaces the standard measurement-based approach (Hamiltonian partitioning + sorted insertion decomposition) with shadow tomography, following the ideas of Ren et al. (Ref [24] in the Q-SENSE paper) applied to the Q-SENSE framework.

Background: Q-SENSE PT Method

The Q-SENSE (Quantum SENiority-based Subspace Expansion) algorithm constructs a subspace Hamiltonian matrix $\mathbf{H}$ from orthogonal basis states built using seniority operators. Each basis state factorizes into classically tractable parts and a quantum part requiring measurement:

- **Diagonal elements:** $H_{ii} = \langle \phi_i^*(Q) | h_{ii} | \phi_i(Q) \rangle$ (linear expectation values)
- **Off-diagonal elements:** $H_{ij} = \langle \phi_i^*(Q) | h_{ij} | \phi_j(Q) \rangle$ (bilinear, between different states)

The ground-state energy is obtained by diagonalizing the subspace Hamiltonian, and the sampling cost metric $\epsilon^2 * M$ quantifies measurement efficiency.

Original Approach

(main_sampling_cost_PT_parallel.py)

The original PT implementation:

1. **Diagonal elements:** Computes $\langle \psi | H | \psi \rangle$ exactly via sparse matrix-vector multiplication.
2. **Off-diagonal elements:** Uses the **extended swap test** formalism (Eq. 27-28 in the paper), which maps $\langle \phi_\mu | h | \phi_\nu \rangle$ to an expectation value of a composite state $|\Phi\rangle = (|0\rangle|\phi_\mu\rangle + |1\rangle|\phi_\nu\rangle)/\sqrt{2}$. The Hamiltonian is augmented with an ancilla qubit via `XorY_augment`.
3. **Sampling cost:** Partitions the effective Hamiltonian into fully-commuting (FC) fragments via the sorted insertion (SI) algorithm, then computes variance-based sampling cost from `variance_of_decomp`.
4. **Constant-term optimization:** Shifts off-diagonal Hamiltonians by `ij_shift = 0.5*(H_ii + H_jj)` to minimize estimator variance (Section III.B.3 of the paper).

Shadow Tomography Approach

(main_sampling_cost_PT_ST.py)

The shadow tomography variant replaces the fragment-based measurement strategy with

classical shadows. The key modifications are organized into four phases:

Phase 1: Pre-computation and State Registration

- Extracts quantum states $|\phi_i\rangle$ for all matrix elements using the same Q-SENSE factorization pipeline (seniority tapering, classical/quantum partitioning).
- Registers **unique quantum states** to avoid redundant shadow collection when the same state appears in multiple matrix elements.
- Pre-computes all tapered Hamiltonians H_Q and caches them (`HQ_cache`).

Phase 2: Shadow Collection (Parallel)

- For each unique quantum state, collects `NUM_SHADOWS` classical shadows using random Clifford measurements.
- Shadows are stored in **compact form**: (Clifford tableau, measurement outcome) pairs, which are $O(n^2)$ per shadow instead of $O(4^n)$ for full unitary matrices.
- For off-diagonal elements, also prepares $|0\rangle +/ - |\psi\rangle$ **superposition states** and collects their shadows:

$$\begin{aligned} \bigcirc \quad |+_i\rangle &= (|0\rangle + |\psi_i\rangle) / \sqrt{2} \\ \bigcirc \quad |-_i\rangle &= (|0\rangle - |\psi_i\rangle) / \sqrt{2} \end{aligned}$$

- Shadow collection is parallelized across all available CPU cores via `multiprocessing.Pool`.

Phase 3: Matrix Element Estimation

- **Diagonal elements** (H_{ii}): Uses `ShadowEstimator.estimate_linear()` -- averages $\text{Tr}(H * \rho_{\text{hat}})$ over shadow snapshots. This is the standard linear shadow estimator.
- **Off-diagonal elements** (H_{ij}): Uses a **bilinear shadow estimation** approach with $|0\rangle +/ - |\psi\rangle$ superpositions:

$$H_{ij} = 2 * (\text{Tr}(\rho_i^+ @ \rho_j^+ @ H) + \text{Tr}(\rho_i^- @ \rho_j^- @ H))$$

where ρ_i^+ and ρ_i^- are density matrices of the $+/-$ superposition states. This is computed via `ShadowEstimator.estimate_bilinear()` which uses vectorized einsum operations for efficiency.

- Estimation is parallelized via `multiprocessing.Pool`.

Phase 4: Exact Comparison

- Computes exact matrix elements via direct sparse matrix-vector products for validation.
- Outputs element-by-element comparison (ST estimate vs exact), error statistics, and energy comparison.

Shadow Tomography Module (`shadow_tomography.py`)

Provides the core shadow tomography primitives:

ClassicalShadow

- Manages shadow collection for a quantum state.
- `get_shadow_snapshot()`: Reconstructs a single shadow density matrix via the inverse channel: $\rho_{\text{hat}} = (2^n + 1) (U^{\text{dag}} |b\rangle\langle b| U) - I$

ShadowEstimator

- `estimate_linear()`: Estimates $\text{Tr}(O * \rho)$ from shadows -- standard linear estimator. Used for diagonal matrix elements.
- `estimate_pauli_expectation()`: Optimized estimator for Pauli string observables (lower variance than general observable estimation).
- `estimate_bilinear()`: Estimates $\text{Tr}(\rho_i @ O @ \rho_j)$ from two sets of shadows. Uses vectorized numpy/einsum for batched trace computation over all shadow pairs. Supports median-of-means for robust estimation. Used for off-diagonal matrix elements.
- `u_statistics_order3()`: Order-3 U-statistics estimator for pure state density matrices.

Simulation Helpers (in `main_sampling_cost_PT_ST.py`)

- `simulate_shadow_collection()`: Simulates quantum shadow collection classically -- generates random Cliffords, applies to state, samples measurement outcomes. Returns compact (Clifford, outcome) tuples.
- `reconstruct_shadow()` / `reconstruct_shadows()`: Converts compact shadow tuples back to full density matrices for estimation.

Key Differences from Original

Aspect	Original (PT Parallel)	Shadow Tomography (PT ST)
Diagonal estimation	Exact sparse multiplication	Linear shadow estimator
Off-diagonal estimation	Extended swap test + FC fragments	Bilinear shadow estimation with
Sampling cost model	Variance from fragment decomposition	Analytical variance scaling: $O(2^n / N_{\text{shadows}})$ diagonal, $O(2^{(2n)} / N_{\text{shadows}}^2)$ off-diagonal
Constant-term shift	<code>ij_shift = 0.5*(H_ii + H_jj)</code>	Not used (unshifted Hamiltonian)
Parallelism	<code>joblib Parallel</code> for off-diagonal	<code>multiprocessing.Pool</code> for both shadow collection and estimation
State reuse	No caching	Unique state registration avoids redundant shadow collection
New parameter	N/A	<code>NUM_SHADOWS</code> (3rd CLI argument) controls accuracy vs cost tradeoff

Motivation

Shadow tomography reduces the measurement scaling with respect to the number of basis states from **quadratic to linear** (Ref [24]), since shadows collected for a state can be reused across all matrix elements involving that state. This is particularly beneficial for the PT

method which has a large number of basis states (e.g., 166 for N_2). The tradeoff is that classical post-processing cost may grow, and the variance of bilinear (off-diagonal) estimation scales exponentially with qubit count.